Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)		
Lab exp 12	**Date: 23/10/2026**	**Enrolment No:92400133079**

CODE

```
import cv2
import numpy as np
from google.colab.patches import cv2_imshow

# 1. Load the image in grayscale
image = cv2.imread('/content/IMG_20260315_113925.jpg.jpeg', 0)

# Check whether the image was loaded successfully
if image is None:
    raise FileNotFoundError("Image not found. Check the image path.")

# 2. Convert the image into a binary image
_, binary = cv2.threshold(
    image, 127, 255, cv2.THRESH_BINARY
)

# 3. Define the pattern to be detected
#  1  = Foreground pixel
# -1  = Background pixel
#  0  = Don't care
pattern = np.array([
    [-1,  1, -1],
    [ 1,  1,  1],
    [-1,  1, -1]
], dtype=np.int8)

# 4. Apply the Hit-or-Miss Transformation
result = cv2.morphologyEx(
    binary, cv2.MORPH_HITMISS, pattern
)

# 5. Display the original binary image and the result
print("Original Binary Image")
cv2_imshow(binary)

print("Hit-or-Miss Result")
```

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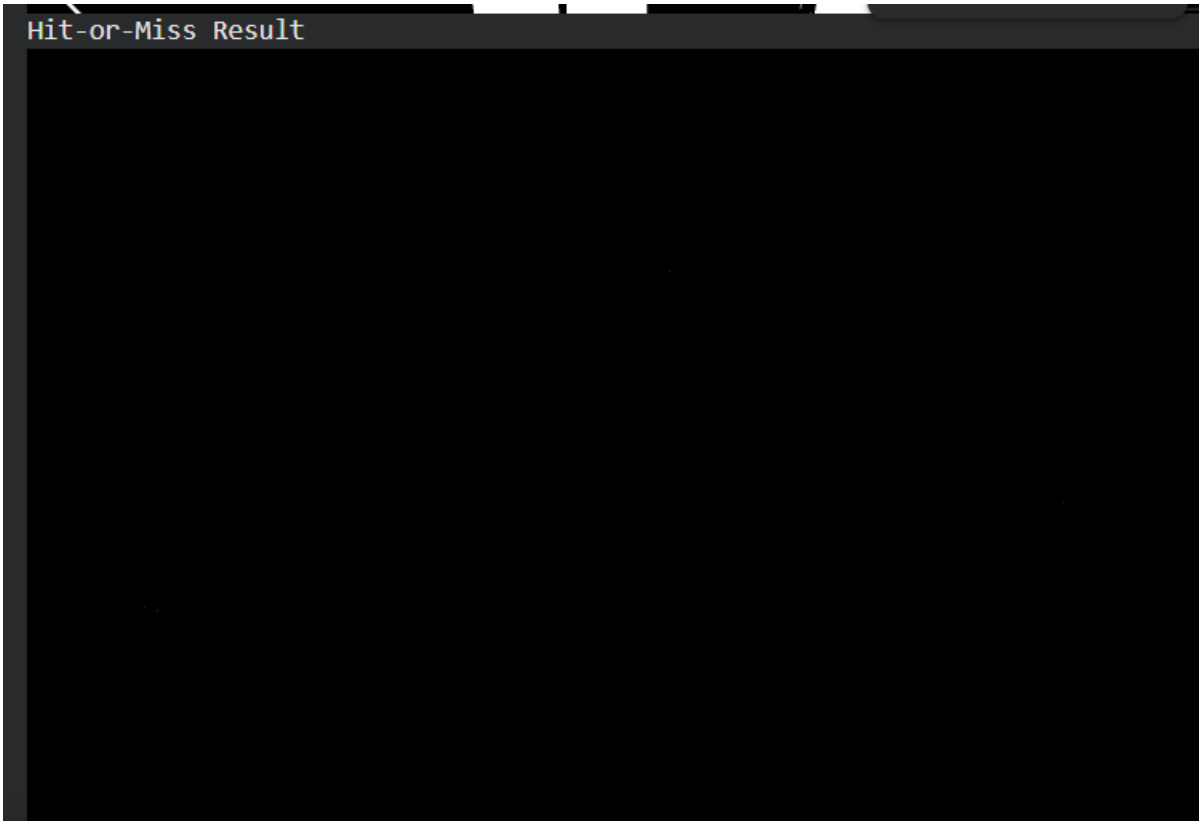
cv2_imshow(result)

6. Close OpenCV windows (not required in Google Colab)
cv2.waitKey(0)
cv2.destroyAllWindows()

OUTPUT



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GITHUB : https://github.com/yugmakwana99/DSIP.git

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